

Historical Olympic Games Analysis (1896–2016)

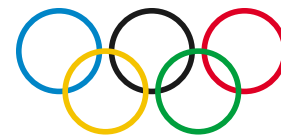
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Problem Statement & Approach



- Problem: Olympic medals are concentrated in a few nations (top 10 hold 60.80%), and National Olympic Committees lack data-driven guidance for funding and talent development.
- How has gender participation evolved since 1896?
- Which nations dominate medals, and how efficiently do they convert entries?
- How do height, weight, and age differ by sport?
- Do nations win by specializing in one sport?

Dataset Overview



269,731

athlete entries

39,772

medals awarded

66

sports

765

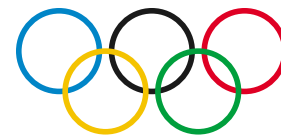
events

1,184

teams

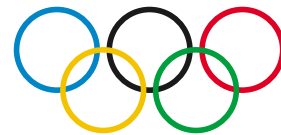
- Period: 1896–2016, Summer and Winter Games
- Location: Global, with athletes from 230 National Olympic Committees and host cities worldwide

Key Terms

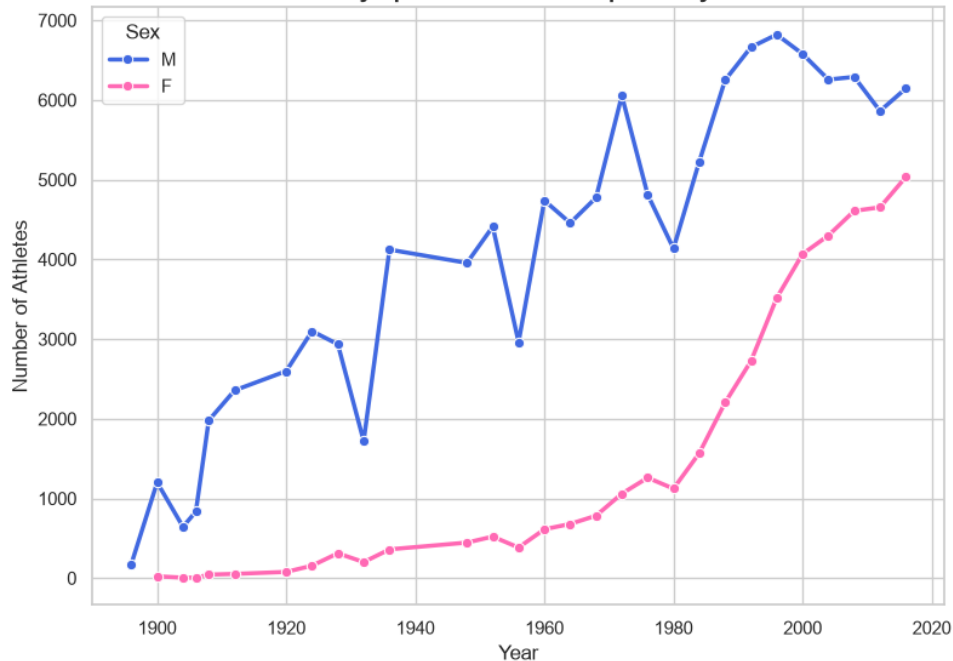


- NOC / Region: National Olympic Committee, used to group athletes by country
- Entry: one athlete competing in one event at one of the Games
- Medalist: an athlete who won Gold, Silver, or Bronze
- Medal conversion rate: $\text{medals} \div \text{entries}$, as a %
- Specialization: share of a nation's medals won in a single sport

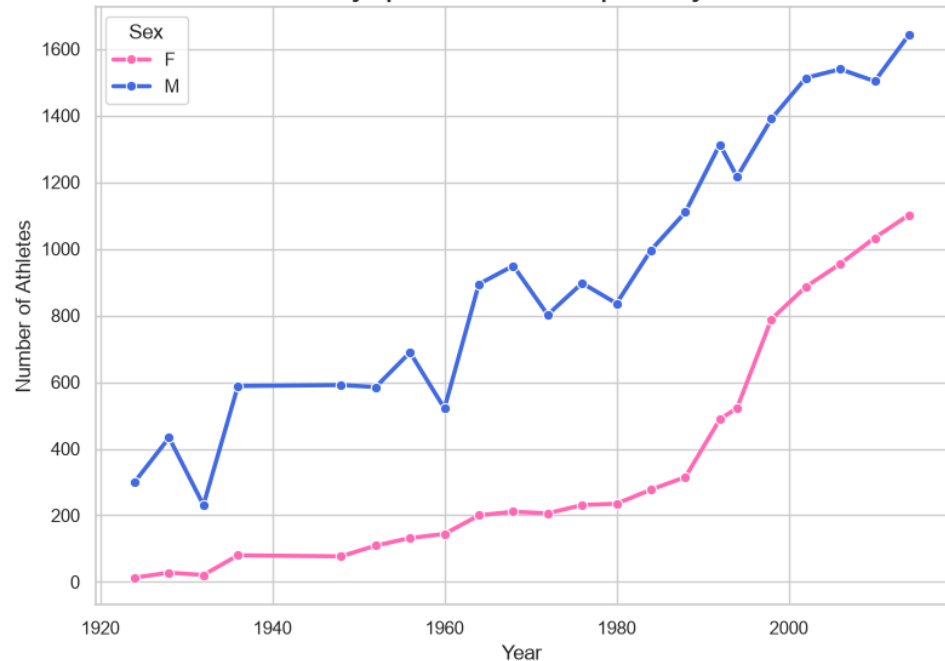
The Rise of Female Olympic Athletes



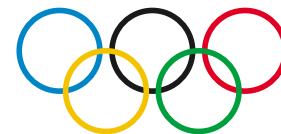
Summer Olympics: Athlete Participation by Gender



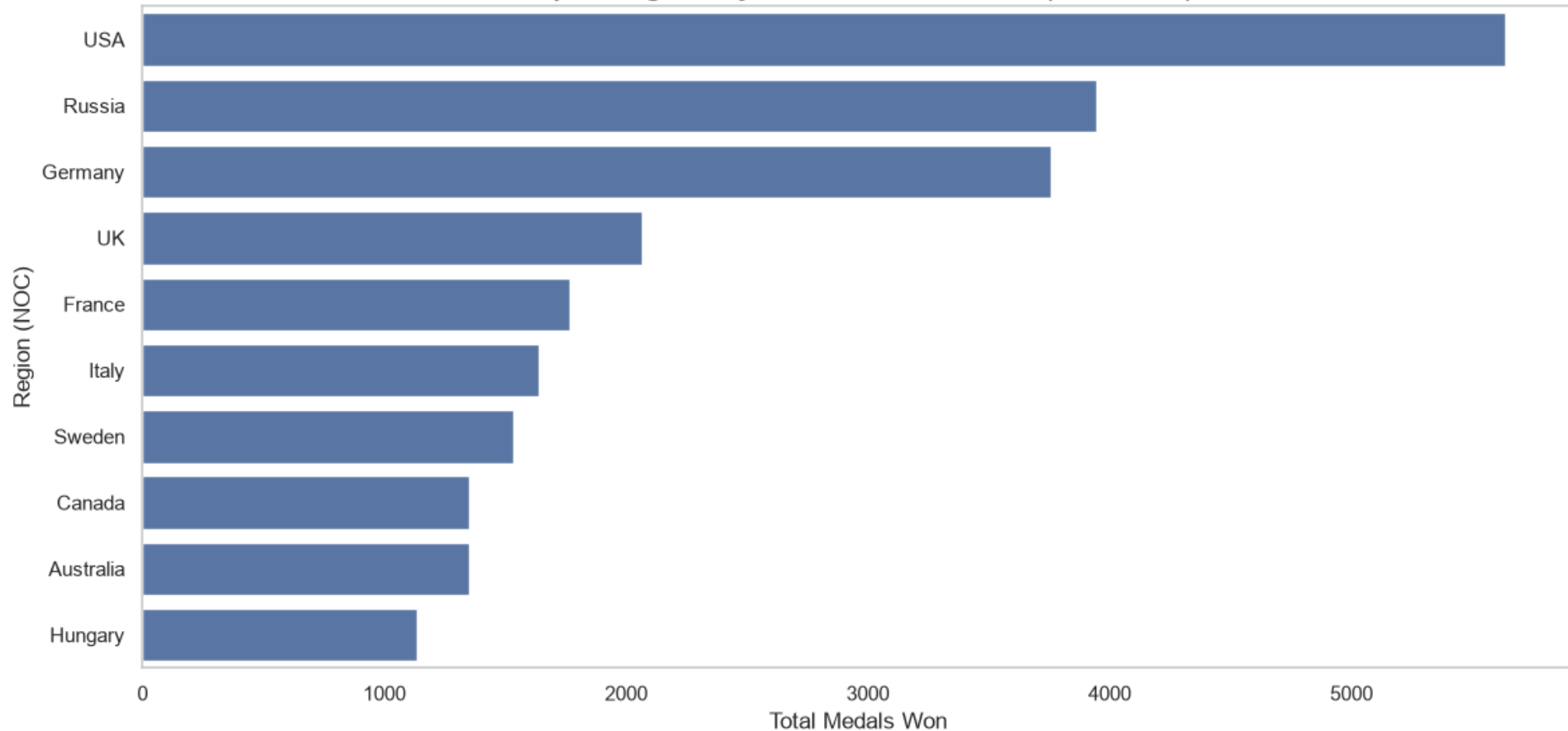
Winter Olympics: Athlete Participation by Gender



Global Medal Distribution



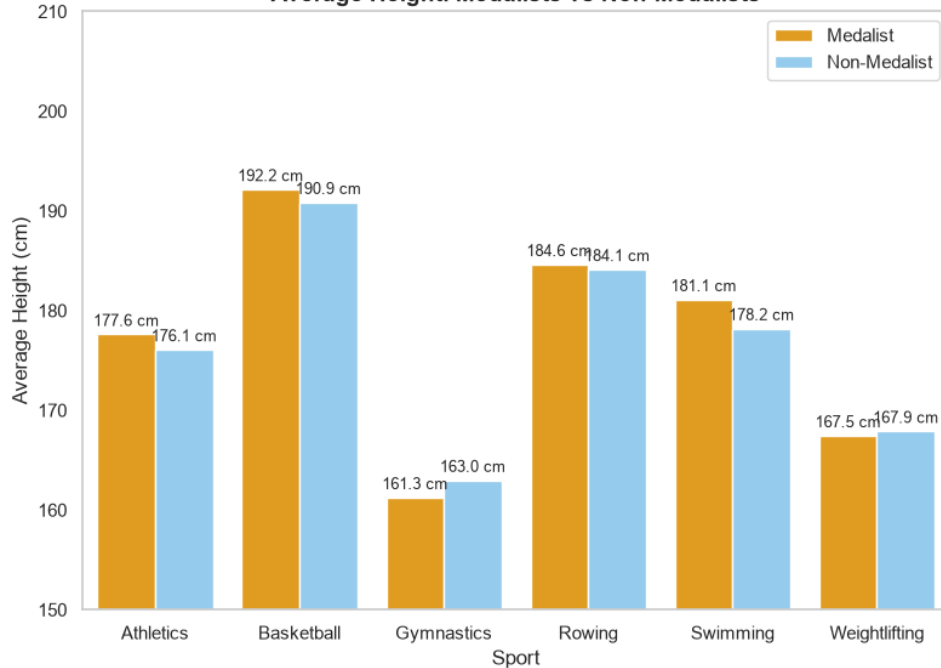
Top 10 Regions by Total Historical Medals (1896–2016)



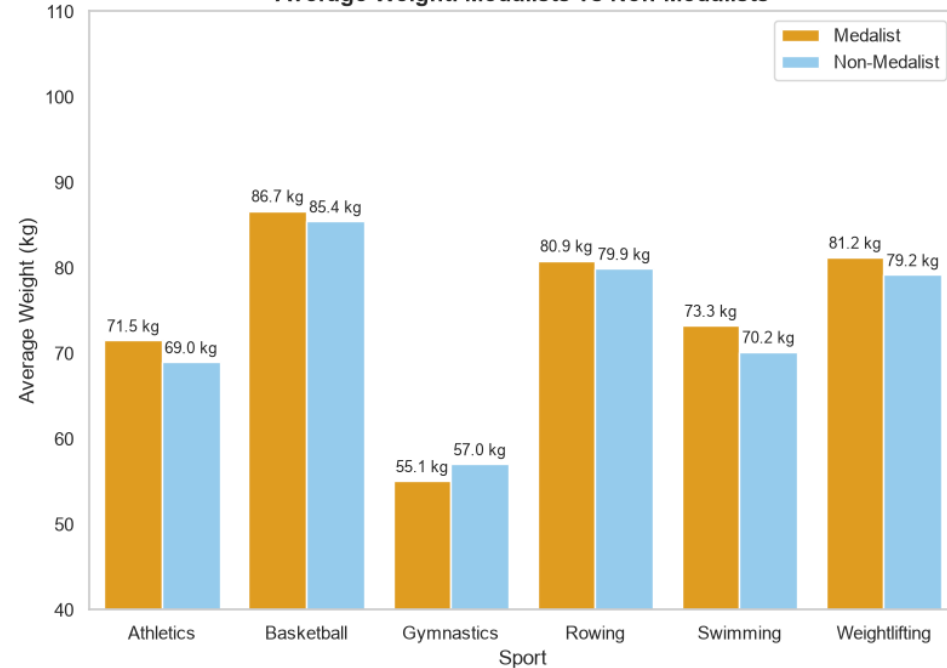
How Physical Profiles Vary by Sport



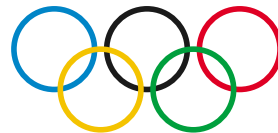
Average Height: Medalists vs Non-Medalists



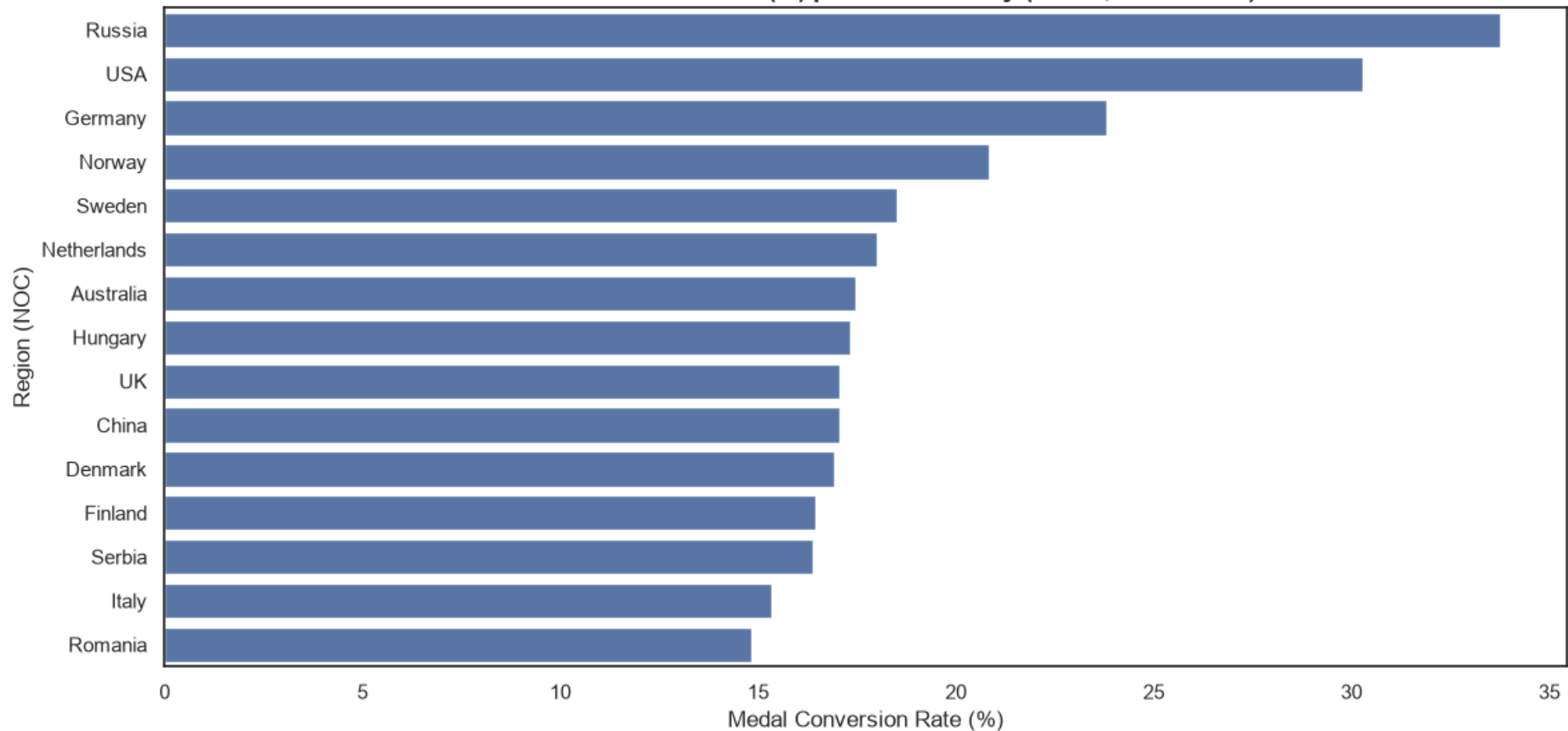
Average Weight: Medalists vs Non-Medalists



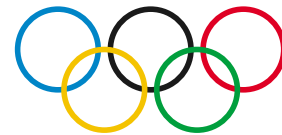
Entry-to-Medal Conversion Rates by Delegation



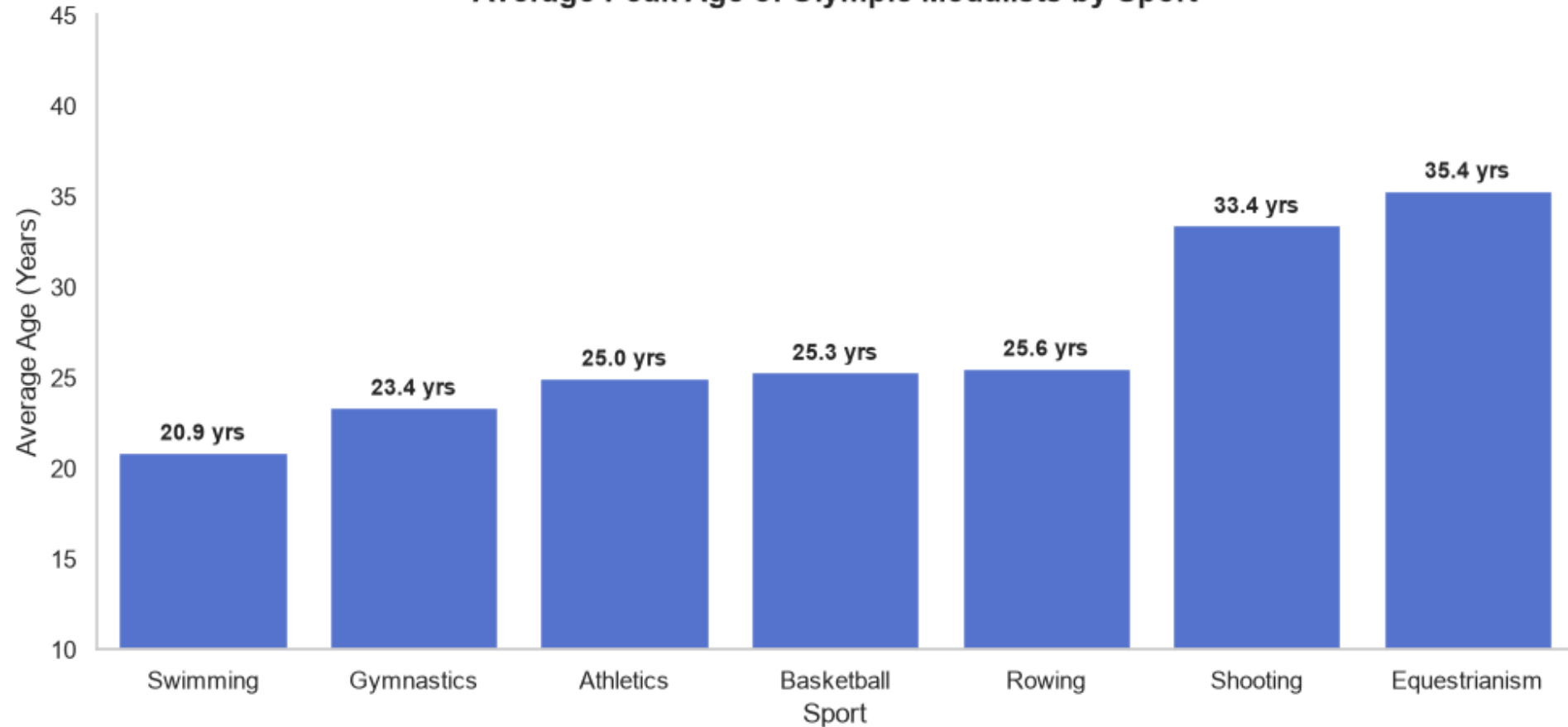
Medal Conversion Rate (%) per Athlete Entry (Min. 3,000 Entries)



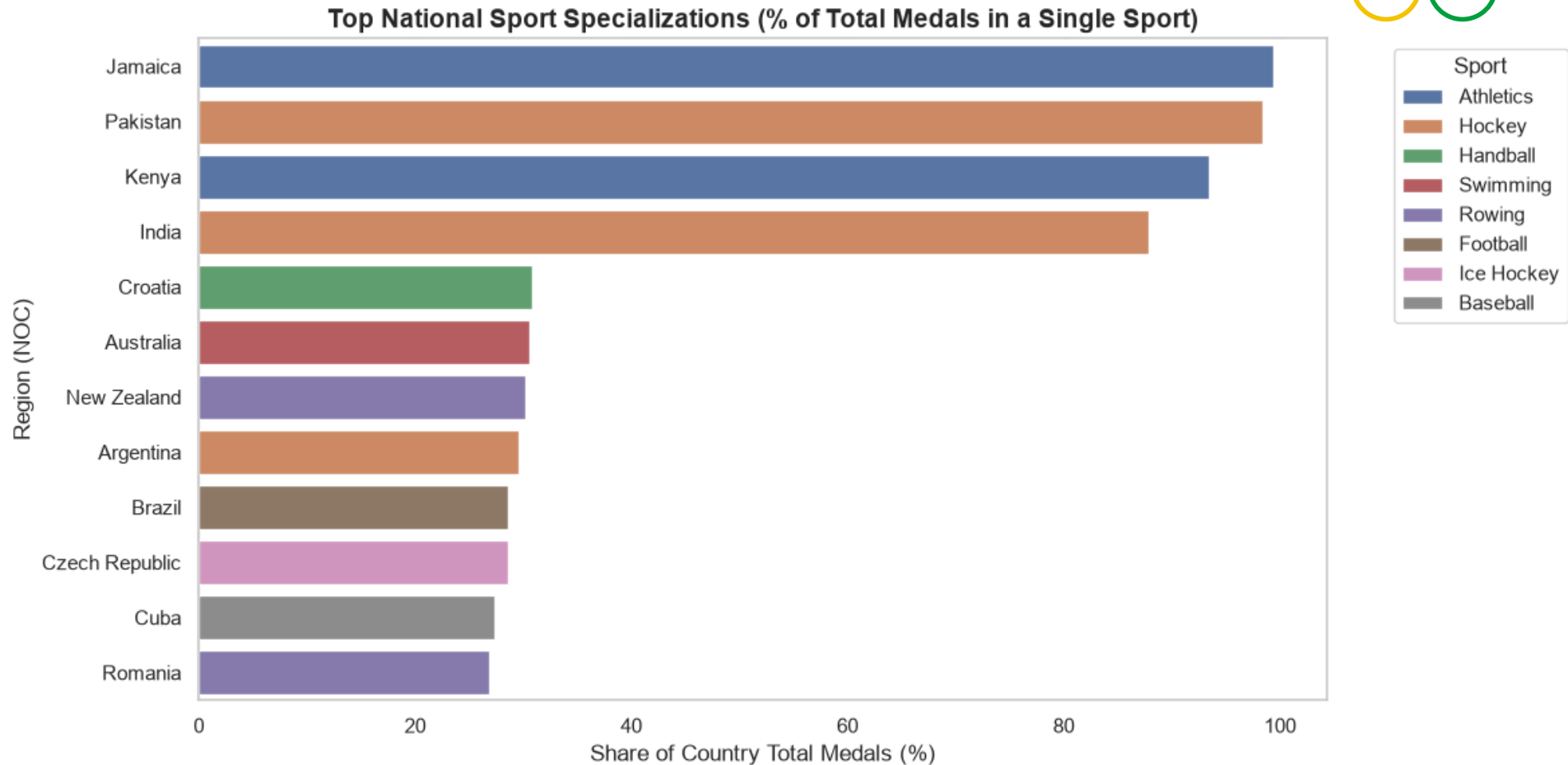
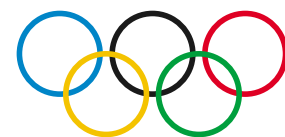
When Athletes Reach Their Olympic Peak



Average Peak Age of Olympic Medalists by Sport



How Focused Delegations Capture Medals

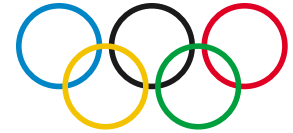


Conclusion



- Female participation grew from 0% in 1896 to over 45% in 2016
- The top 10 nations hold 60.80% of all medals
- Physical profile and peak age differ sharply by sport
- Several nations win almost entirely through one sport

Recommendations



- Match talent to sports whose medalist profiles fit (e.g., 192 cm in Basketball vs 161 cm in Gymnastics)
- Time athlete development to each sport's peak age window (early 20s vs 30s)
- Smaller delegations can concentrate resources in one proven sport, as Jamaica and Kenya do in Athletics
- Keep expanding women's events, given participation nearing parity